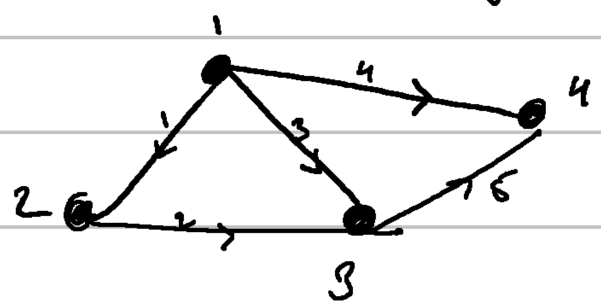
Lecture 12

⇒ Applications of linear algebra

- Graph

Nodes & edges



4×4 matrix

Incidence matrix → Every row is an information about edge
cols are nodes.

$$\begin{array}{c|c} E \backslash N & \begin{matrix} 1 & \ \ 2 & \ \ 3 & \ \ 4 \end{matrix} \\ \hline \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \end{array}$$

rank $(A) = 3$

Interesting point →

edges ① ② ③ form a loop and this can be deduced from incidence matrix by seeing that rows 1-3 are not linearly independent.

↳ Lets say A is the previous incidence matrix & $x_1 \ldots x_4$ are potentials at nodes.

$$A x = 0$$

↳ computes the potential diff.

$$\begin{bmatrix} x_2 - x_1 \\ x_3 - x_2 \\ \vdots \\ \vdots \end{bmatrix}$$

Basis for the null space →

$$\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \qquad N(A) = \lambda \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

↳ Basically what this tells us is that no current (nullspace) when the potentials are all the same in the network.

- Grounding a node is equivalent to setting that free variable value to zero.

$$\boxed{\text{rank}(A) = 3}$$

$N(A^T) \Rightarrow \dim(5-3) = 2$ dimensional

$$\begin{bmatrix} -1 & 0 & -1 & -1 & 0 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \qquad A^T y = 0$$

$$\hookrightarrow \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 1 & 0 & -1 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} -F \\ I \end{bmatrix} \qquad \begin{bmatrix} -1 & 1 \\ -1 & 1 \\ 0 & -1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} -1 & 1 \\ -1 & 1 \\ 1 & 0 \\ 0 & -1 \\ 0 & 1 \end{bmatrix}$$

$$\boxed{\dim(N(A^T)) = 2}$$

special sol^n & basis

$x_1, x_2 \;--\; x_n$ $\rightarrow$ potentials $[x]$

$\downarrow Ax$

$[x_1 - x_2, \;\cdots\cdots \;]$ potential diffs.

ohm's law $\downarrow C$

Let $y_1 --- y_5$ be the edge currents.

$A^T y = 0$ $\hookrightarrow$ net current in a closed circuit is zero !!

Kirchoff's law.

• col space of $A^T \rightsquigarrow \dim(C(A^T)) = 3$
$= \underline{\underline{\text{rank}}}$

Lets say we take ① ② & ④ as they are linearly independent and they can form a basis for the $\underline{\underline{C(A^T)}}$

$\rightsquigarrow$ $\underline{\underline{\text{Tree}}}$ $\rightarrow$ a graph with no loops.

\# $\underbrace{\dim}_{\downarrow}(N(A^T)) = m - r$

$\# \text{ loops} = \# \text{ edges} - (\# \text{ nodes} - 1)$

$$\boxed{\# \text{ nodes} + \# \text{ loops} = \# \text{ edges} + 1}$$

⇒ This is also called euler's formula. [Topology]

Summary

Ax → potential difference matrix

CAx → c → physical constant conductance.
→ currents.

$A^T c A x = f$ → external current sources

→ $A^T A$ is always a symmetric matrix.